

Divide and Conquer Example

You are given an array A of n numbers. A[i] is the price of a specific stock on day i. Your goal is to figure out the best days to buy and sell that stock in order to maximize profit. That is, you must find two days i, j such that i < j, and A[j] - A[i] is maximized. Brute-force algorithm for this problem has a complexity of $O(n^2)$, but we are seeking for a more efficient algorithm.

(a) Here is a divide and conquer algorithm for this problem:

1. If array has at most 2 elements, solve the problem trivially, return.

2. Solve the problem in first n/2 elements (recursively). Let p_1 be the maximum profit.

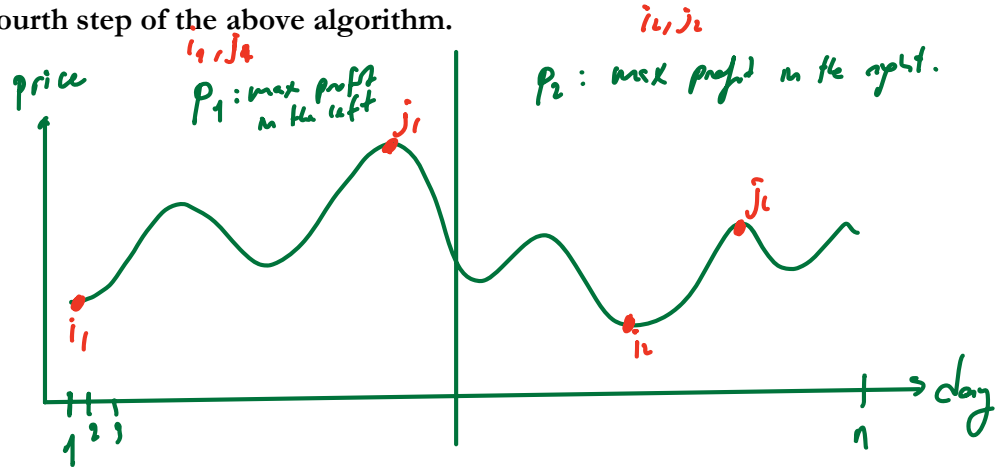
3. Solve the problem in last n/2 elements (recursively). Let p_2 be the maximum profit.

4. $p_3 \leftarrow \max\{A[\frac{n}{2}..n-1]\} - \min\{A[0..\frac{n}{2}-1]\}$ ✓

$p_3 \leftarrow A[j_2] - A[i_1]$ ✗ → not correct.

5. Return the maximum of p_1, p_2 and p_3.

Fill the fourth step of the above algorithm.



$T(n) = 2T(\frac{n}{2}) + f(n)$ $\quad f(n) \in \Theta(n)$

$T(n) \in \Theta(n \log n)$